GenDR

A Generic Deep Realizer with Complex Lexicalization

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NLG

Data ➡ Assessment ➡ Planning ➡ Realization

GenDR

Semantics ➡ Syntax ➡ Morphology ➡ Text

Surface realizers

Salience-driven realization

Which meaning is the most salient?

‘small’ —1→ ‘cat’ ←1— ‘lie’ —2→ ‘sofa’ ←1— ‘red’

≈ small(e_1, x) ∧ cat(x) ∧ lie(e_2, x, y) ∧ sofa(y) ∧ red(e_3, y)

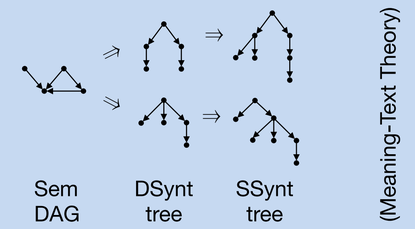
The cat lying on the red sofa is small.

The small cat is lying on the red sofa.

The sofa on which the small cat is lying is red.

Semantics-syntax interface

Arborization



Lexicalization

- Simple (lexemes)
- Template-based (full idioms)
- Bound (collocations)
- Class-based (numbers, etc.)
- Grammatical (function words)
- Fallback (unknown words)

Lexical resources

Lexical dictionary

```
owe {
  dpos=V spos=verb
  gp = { // X owes Y to Z
    1=I 2=II 3=III  // trivial diathesis
    I={dpos=N rel=subj}
    II={dpos=Num rel=dobj}
    III={dpos=N rel=iobj prep=to} } }

debt {
  dpos=N spos=noun
  gp = { // X's debt of Y to Z
    1=II 2=I 3=III  // special diathesis
    I={dpos=Num rel=ncomp prep=of}
    II={dpos=N  rel=poss  case=GEN}
    III={dpos=N rel=ncomp prep=to} }
  lf={name=Oper1  value=have}
  lf={name=Oper13 value=owe}
  lf={name=Func2  value=amount_v_1}
  lf={name=Func2  value=stand_v_2}
  lf={name=Func2  value=total_v} }
```

Semantic dictionary

```
owe {
  lex=owe
  lex=debt }
```

Lexical coverage

English	1,500
French	1,500
Lithuanian	180
Persian	60

Lexical functions meta-dictionary

```
Oper1 {  // e.g., X has a debt
  dpos = V
  gp = { 1=I L=II } }

Oper13 { // e.g., X owes a debt to Z
  dpos = V
  gp = { 1=I L=II 3=III } }

Func2 {  // e.g., the debt amounts to Y
  dpos = V
  gp = { L=I 2=II } }
```

Arg 1 → Actant I
Base → Actant II

Lexical Functions (LFs)

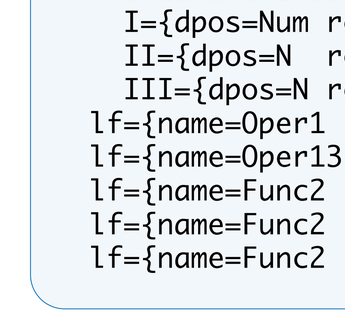
Collocations tend to be instances of recurrent patterns across languages, e.g., *strong preference*, *gravely ill*, *intense flavour* and *win hands down* are all instances of the same pattern where a base is intensified by a syntactic modifier (the collocate).

What defines a collocation is the special relationship that exists between the base and the collocate it selects, which is modeled as a function. Intensification collocations are modeled with the *Magn* function: *Magn*(*preference*)=*strong*, etc.

Over the years, around 60 basic LFs have been identified, that combine to form a large number of complex LFs. GenDR implements ~26,000.

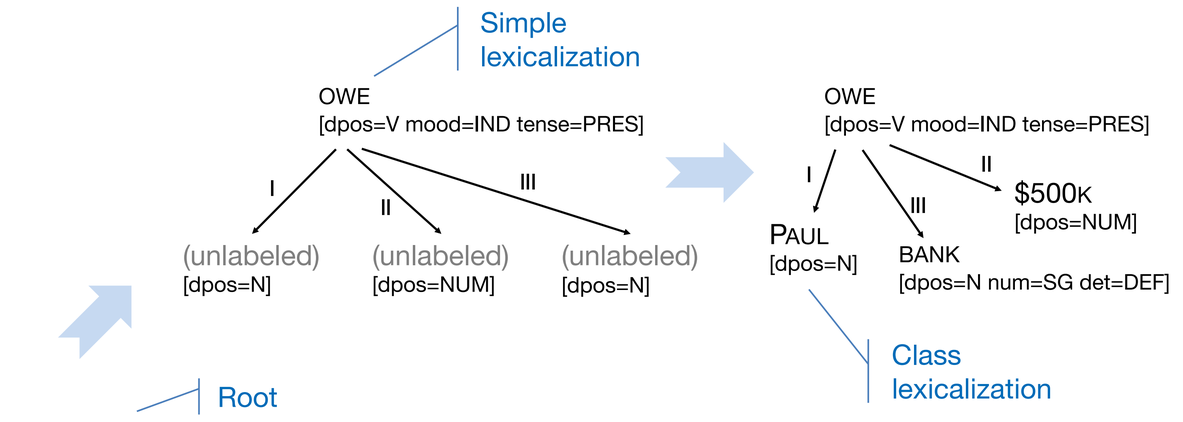
Step-by-step example

Semantics

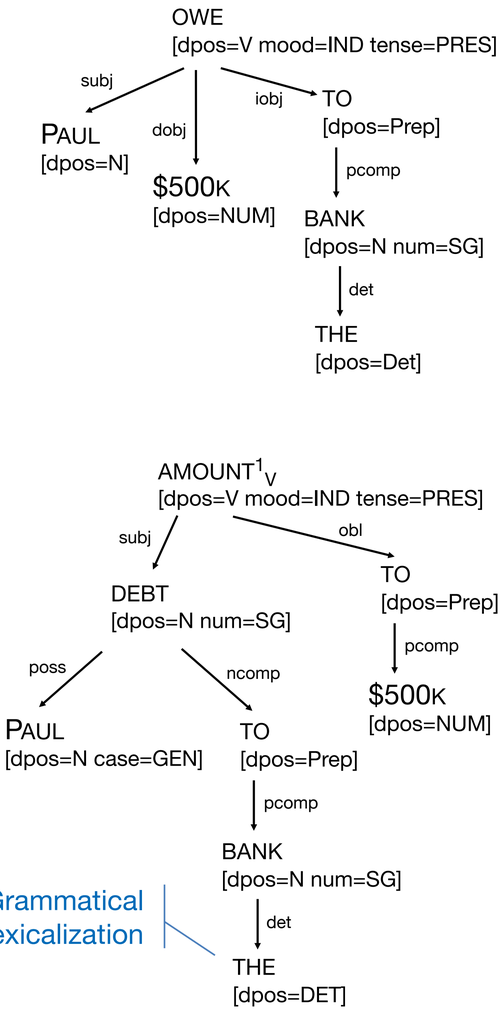


Deep syntax

Paul owes $500k to the bank.



Surface syntax



Paul’s debt to the bank amounts to $500k.

DPOS	deep part of speech
GP	government pattern
LF	lexical function

See the paper for references